A Head Camera Is (Almost) All You Need

An electronics-free UMI glove labelled and observed by one moving camera

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Why a head camera only

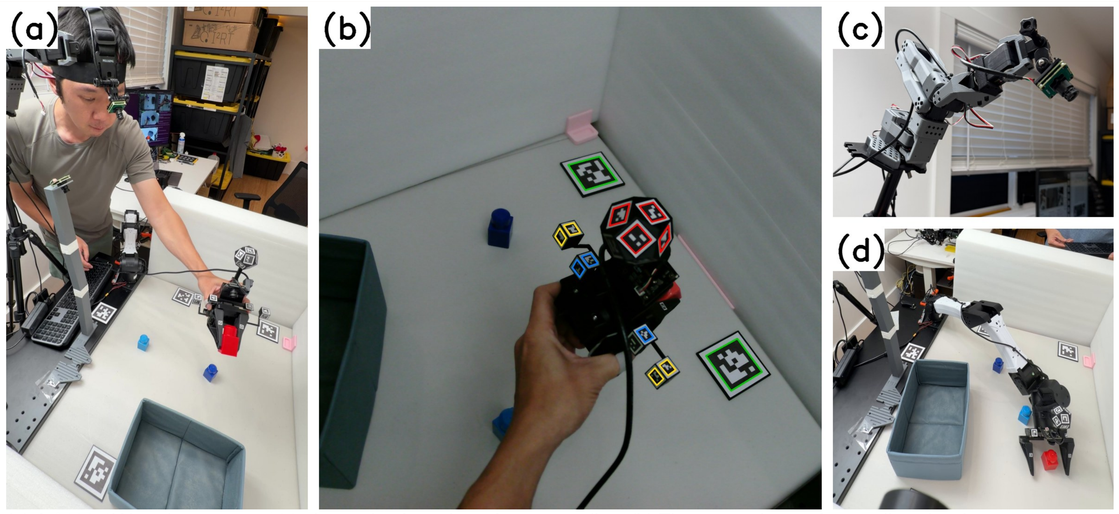
- Wrist devices need a **second stream** to track the gripper: SLAM, an IMU, a controller or an external camera.
- **Gaze leads the hand**: the gripper stays in view, at arm's length, with the goal.
- The frames the policy sees are the frames that **carry the labels**.

No SLAM, no IMU, no wrist camera, no cables.

$144 glove, camera, strap

103 g on the head

One head camera: the policy's observation and the pose labels



(a) A $122 4K head camera, a $22 electronics-free glove, four paper tiles. (b) The camera's own frame: tiles give the camera pose, the ball the gripper pose, the jaw tags the opening. (c) At deployment the same camera rides a $122 six-joint arm (d) beside the robot.

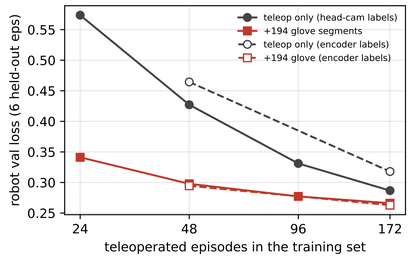
194 glove demos stretch a small teleop set

Eight ACT policies, joint-space actions. Rollouts: 25 placements, one trial each, 60 s. Every co-trained mix uses the same 194 glove demos.

10/25 48 teleop episodes

17/25 48 teleop + 194 glove

1 : 2–3 teleop episode : glove demos



9 placements won to 2 lost ($p = 0.065$); loss 0.464 → 0.295, the level of 96–172 teleop episodes. On the teleop-only curve the glove set is worth **65–100 teleop episodes**, an exchange rate of 2–3 glove demos per teleoperated one. Added to all 172 it still helps: 18/25 → 20/25, loss 0.318 → 0.263.

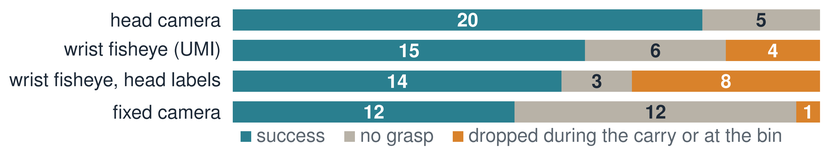
Head-camera labels can replace the encoders

0.318 teleop labelled by encoders

0.328 teleop labelled by the head camera

Same 172 episodes, same held-out targets: 3 % of loss. On the robot 18/25 vs 15/25, within noise ($p = 0.58$).

Where each view fails



Offline the four co-trained views sit within 0.011 of each other. On the robot, wrist-camera policies **drop the block mid-carry**, every time while the fisheye looked away from the bin. The head camera and the fixed camera keep the bin in frame and almost never drop; they fail by not grasping. Caveat: the teleoperator watched the head-camera preview while demonstrating.

Label accuracy

- **Camera pose** from the paper tiles, refit from the footage to 0.5 px.
- **Gripper pose** by PnP on a 15 mm ArUco dodecahedron.
- **Jaw width** from printed tails on the jaws.

6.5 mm tool point

1.6° orientation

3.5 % jaw width, of stroke

Median error against the robot's encoders; a label on 93–100 % of frames.

Collection

2–3× demos per minute vs teleop

198 demos in four sessions

6–8 s per demo; teleop 14 s

0 robot time

Continuous recording, episodes cut by a keypress; 0.7–7.4 % of frames dropped for tracking.

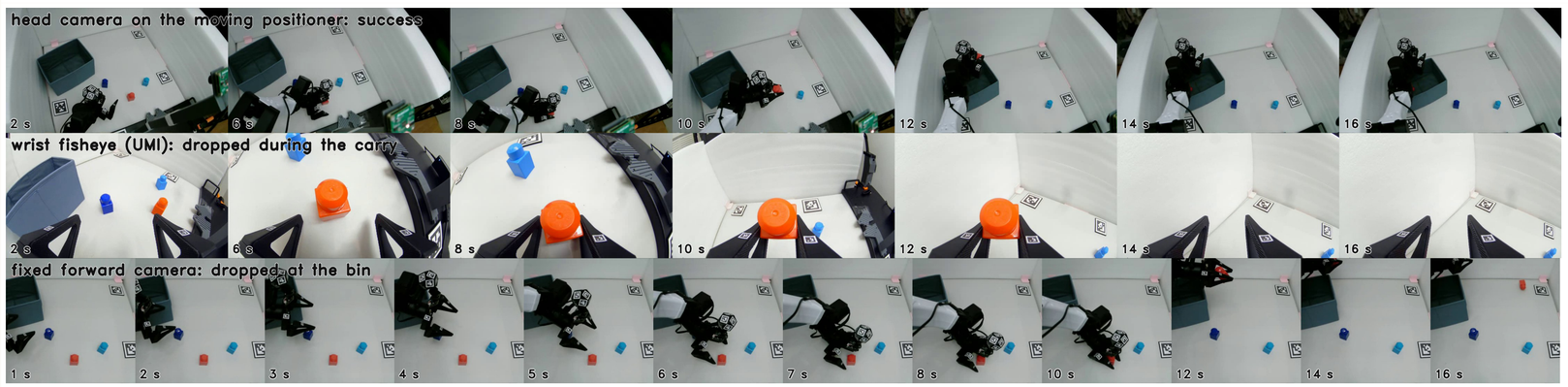
Deployment

- The same camera on a $122 six-joint arm beside the YAM.
- It follows the gripper **from the robot's encoders alone**, in teleoperation and in rollouts, so the policy sees what the collector saw.

46 mm head pose vs the collector

5.4° static camera: 87 mm, 7.0°

One placement, seen by each policy



Placement 8, each row the frames that policy itself saw. The head camera keeps gripper and bin in one frame through the carry. The wrist fisheye sees only the block and the wall from the grasp on, and drops the block before the bin comes into view. The fixed camera sees the whole scene, reaches the bin, and drops the block just outside it.

It is the view, not the labels

- Same wrist view, labels from the fixed camera or from the head camera: **15/25 vs 14/25**, no change.
- Same head labels, wrist view or head view: **14/25 → 20/25**, and the drops disappear.

Limits and next

One task, one collector, one tile layout; teleoperation watched the head-camera view. Glove-only policies do not transfer: co-training is required. Next: bimanual, tile-free in the wild.